TEXORA · LECTURE SHEET

Hydroxy Compounds

Alcohols and phenols: classification, preparation, hydrogen bonding and acidity, the characteristic reactions — Lucas test, oxidation ladder, dehydration, esterification — and phenol's special chemistry, from bromine water to Kolbe and Reimer–Tiemann.

CHE-1205 · CHEMISTRY-II · TOPIC 2 · LEVEL 1 TERM 2

1. Classification

Hydroxy compounds carry the –OH group. When –OH sits on an alkyl chain they are **alcohols** (R–OH); on an aromatic ring directly, **phenols** (Ar–OH) — a small structural difference with large chemical consequences (Section 4). Alcohols classify two ways:

BY –OH POSITION	DEFINITION	EXAMPLE
Primary (1°)	–OH carbon bonded to ≤ 1 other carbon	Butan-1-ol
Secondary (2°)	–OH carbon bonded to 2 carbons	Butan-2-ol
Tertiary (3°)	–OH carbon bonded to 3 carbons	2-Methylpropan-2-ol

By –OH count: monohydric (ethanol), dihydric (ethane-1,2-diol, "glycol"), trihydric (propane-1,2,3-triol, glycerol — the backbone chemistry of fats and of alkyd resins in textile finishing).

2. Preparation

- **Hydration of alkenes:** acid-catalysed addition of water across C=C, Markovnikov orientation (ethene → ethanol industrially, with H_3PO_4 at high T, p).
- **Fermentation:** glucose → 2 C_2H_5OH + 2 CO_2 (yeast enzymes, zymase) — the oldest chemical industry.
- **Reduction of carbonyls:** aldehydes → 1° alcohols, ketones → 2° alcohols ($NaBH_4$, $LiAlH_4$, or H_2/Ni).
- **Grignard route:** R–MgX + a carbonyl compound builds alcohols with a lengthened carbon skeleton (formaldehyde → 1°, other aldehydes → 2°, ketones → 3°).
- **Hydrolysis of alkyl halides** with aqueous KOH; **phenol** industrially from cumene (cumene hydroperoxide cleavage → phenol + acetone) or by fusing benzene sulphonate with NaOH.

3. Properties and reactions of alcohols

The –OH group hydrogen-bonds, so alcohols boil far above comparable hydrocarbons (ethanol 78 °C vs propane −42 °C) and small alcohols mix with water in all proportions; solubility fades as the hydrocarbon part grows (butanol ≈ 8 g per 100 mL). **Acidity** is feeble: ethanol's pKa ≈ 16, weaker than water — yet visible: sodium dissolves in ethanol evolving hydrogen (more gently than in water).

REACTION	BEHAVIOUR — AND WHAT IT DISTINGUISHES
Lucas test (conc. $HCl/ZnCl_2$)	3°: immediate turbidity; 2°: turbidity in ≈ 5 minutes; 1°: none at room temperature — the standard 1°/2°/3° test
Oxidation	1° → aldehyde → carboxylic acid; 2° → ketone (resists further); 3° resists — no α-H path under normal conditions
Dehydration (conc. H_2SO_4, heat)	Alkene formed; ease 3° > 2° > 1°; the more substituted alkene dominates (Saytzeff's rule): butan-2-ol → mainly but-2-ene
Esterification	R–OH + R′–COOH ⇌ ester + H_2O (acid catalyst) — the route to polyester's monomers begins here
Iodoform test (I_2/NaOH)	Yellow CHI_3 precipitate from CH_3–CHOH– structures: ethanol and 2° alcohols with a CH_3 on the carbinol carbon; 1-propanol and methanol fail

4. Phenol — the hydroxy compound that behaves like an acid

Phenol (pKa ≈ 10) is a million times stronger an acid than ethanol. The reason is resonance: the phenoxide ion delocalises its negative charge into the ring (three further contributors), while phenol itself is less stabilised — the equilibrium shifts right. Consequences: phenol dissolves in NaOH (ethanol does not) though it is still too weak to decompose $NaHCO_3$ (no effervescence — the quick test separating phenols from carboxylic acids, pKa ≈ 5).

- **$FeCl_3$ test:** violet coloration — the instant classroom test for phenols.
- **Bromine water:** instantly decolourised with a white precipitate of **2,4,6-tribromophenol** — the –OH activates the ring so strongly that no Lewis-acid catalyst is needed (contrast benzene, which needs $FeBr_3$ and stops at mono-substitution with $Br_2/FeBr_3$).
- **Nitration:** dilute HNO_3 gives o- and p-nitrophenol; concentrated HNO_3/H_2SO_4 gives **picric acid** (2,4,6-trinitrophenol) — a strong acid (pKa ≈ 0.4) and an explosive, despite being a "phenol".
- **Kolbe–Schmitt:** sodium phenoxide + CO_2 (high p, T) → salicylic acid — the aspirin precursor.
- **Reimer–Tiemann:** phenol + $CHCl_3$/NaOH → salicylaldehyde (o-hydroxybenzaldehyde), an electrophilic substitution by dichlorocarbene.

Acidity order to remember: picric acid (0.4) > carboxylic acids (≈ 5) > phenol (10) > water (15.7) > ethanol (16). Electron-withdrawing ring substituents strengthen phenols; electron-donating groups weaken them.

5. Key points

- Alcohols: 1°/2°/3° by the carbinol carbon's substitution; phenols have –OH directly on the ring.
- Hydrogen bonding explains high boiling points and short-chain water miscibility.
- Lucas test separates 1°/2°/3° by turbidity speed; oxidation ladder stops at aldehyde/ketone/acid by class.
- Phenol's acidity (pKa ≈ 10) is a resonance effect of phenoxide; it dissolves in NaOH but does not fizz with $NaHCO_3$.
- Phenol signatures: violet $FeCl_3$, instant 2,4,6-tribromophenol with bromine water, Kolbe → salicylic acid, Reimer–Tiemann → salicylaldehyde.

6. Practice

Answer in your own words; brief model answers follow.

1. Three unlabelled bottles hold butan-1-ol, butan-2-ol and 2-methylpropan-2-ol. Describe one test that identifies all three, with the observations.
2. Write the products of oxidising (a) propan-1-ol with acidified $KMnO_4$ to completion, (b) propan-2-ol likewise, and (c) state why 2-methylpropan-2-ol behaves differently.
3. Butan-2-ol is heated with concentrated sulphuric acid. Name the major and minor alkene products and the rule deciding between them.
4. Phenol decolourises bromine water instantly without a catalyst, giving a white solid. Name the solid and explain the ring's high reactivity in terms of the –OH group.
5. Arrange in decreasing acidity: ethanol, phenol, picric acid, acetic acid. Justify the phenol–ethanol gap.
6. Which of these give a positive iodoform test — ethanol, propan-1-ol, propan-2-ol, methanol? State the structural requirement.

Answer key

1. Lucas test (conc. HCl + $ZnCl_2$, room temperature): 2-methylpropan-2-ol (3°) turns turbid immediately; butan-2-ol (2°) turns turbid in about five minutes; butan-1-ol (1°) stays clear. The test runs on the ease of forming the carbocation-like transition state, 3° > 2° > 1°.

2. (a) Propanoic acid (through propanal). (b) Propanone, which resists further oxidation. (c) The tertiary alcohol has no hydrogen on the carbinol carbon, so the usual oxidation pathway is blocked; it survives conditions that oxidise 1° and 2° alcohols.
3. Major: but-2-ene (the more substituted alkene); minor: but-1-ene. Saytzeff's rule — elimination preferentially removes hydrogen from the carbon that gives the more substituted (more stable) double bond.
4. 2,4,6-Tribromophenol. The –OH group donates electron density into the ring by resonance, raising its reactivity toward electrophiles so much that all three activated positions (2, 4, 6) brominate at once; benzene, lacking the activator, needs $FeBr_3$ and brominates once.
5. Picric acid > acetic acid > phenol > ethanol. Phenol beats ethanol because the phenoxide anion is resonance-stabilised over the ring, whereas ethoxide's charge is localised on oxygen; losing a proton is therefore far easier for phenol.
6. Ethanol and propan-2-ol. The test needs a CH_3–CH(OH)– unit (or CH_3CO–, which oxidises to it under the test conditions). Propan-1-ol's carbinol carbon carries an ethyl group instead; methanol has no second carbon at all.
